MS PROJECT REPORT — INITIAL DRAFT (CHAPTERS 1–3)
EMGT 7945 MS Project

Urban Mobility Equity Dashboard:
Mapping Who Gets Left Behind in U.S. Cities

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Date	June 2026

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1. INTRODUCTION AND MOTIVATION

1.1 Background

Transportation is one of the most fundamental determinants of economic opportunity in urban America. A person's ability to reach employment, healthcare, education, and essential services depends on a layered ecosystem of infrastructure: transit frequency and coverage, pedestrian networks, bicycle facilities, and access to electric-vehicle (EV) charging. Each dimension has documented equity gaps across U.S. cities. Transit deserts—areas where transit supply falls far short of the demand of car-free, low-income residents—disproportionately affect minority communities [1]. Walkability is highly racialized, with communities of color consistently inhabiting less walkable neighborhoods [2]. Disadvantaged communities have roughly 64% fewer public EV chargers per capita than non-disadvantaged communities [3], and bicycle infrastructure remains underfunded in lower-income areas [4].

Despite this evidence, the available tools remain siloed. The EPA National Walkability Index, the DOE Alternative Fuels Station Locator, the FTA National Transit Database, and OpenStreetMap bicycle data each measure one dimension in isolation. No open-access tool integrates all four into a single neighborhood-level equity framework—the gap this project fills.

1.2 Problem Statement

Low-income and minority neighborhoods in major U.S. cities likely experience compounded, multi-dimensional mobility disadvantage, scoring poorly not in one dimension but across several simultaneously. The magnitude of this compounding effect, and the specific neighborhoods where it is most severe, have not been quantified in an integrated, publicly accessible format. This directly impedes evidence-based infrastructure-investment decisions at the city and federal levels.

1.3 Research Questions

The project is guided by one central question and four sub-questions:

1. **Central.** To what extent do low-income and minority neighborhoods in major U.S. cities experience compounded disadvantage across transit access, walkability, bicycle infrastructure, and EV charging, and how can this multi-dimensional inequity be measured, visualized, and acted upon?
2. *Measurement.* Can heterogeneous public datasets be integrated into a single, transparent, tract-level composite index that is reproducible and comparable across cities?
3. *Distribution.* Where are the mobility deserts (bottom-quartile MES) and compounded deserts (bottom quartile on ≥ 2 dimensions) within each city?
4. *Equity.* Is the MES associated with median income, racial composition, vehicle access, and tenure, and in what direction and magnitude?
5. *Robustness.* Are conclusions sensitive to the weighting of sub-indices and to the percentile threshold defining a mobility desert?

1.4 Specific Objectives

1. Construct a Mobility Equity Score (MES) by aggregating four mobility sub-indices at the census-tract level across the study cities.
2. Quantify the correlation between low MES and socioeconomic indicators: median income, racial composition, vehicle ownership, and renter share.

3. Identify and rank the most mobility-disadvantaged tracts in each city, distinguishing single-dimension from compounded multi-dimension deserts.
4. Build an interactive, open-access Plotly Dash dashboard for spatial exploration and city comparison.
5. Produce a policy brief with actionable investment recommendations for city transportation departments.

1.5 Significance

Academically, this project fills a documented gap in the mobility-equity literature by providing the first integrated four-dimension index at the census-tract level with explicit, testable methodological choices. Practically, the dashboard translates the analysis for non-technical stakeholders and supports the federal Justice40 mandate to direct infrastructure benefits to disadvantaged communities [7], helping agencies target investment toward the neighborhoods that need it most.

1.6 Organization of the Report

Chapter 2 reviews the literature on transportation equity, the four mobility dimensions, composite-indicator construction, and spatial methods. Chapter 3 specifies the methodology in full. Chapters 4 and 5 (forthcoming) present results and discussion.

2. LITERATURE REVIEW

2.1 Transit Equity and Transit Deserts

Public transit accessibility is both a material and a social phenomenon shaping employment access and quality of life [5]. Aman and Smith-Colin [1] introduced a transit-desert framework quantifying the supply–demand gap at the census-tract level using GTFS data, finding systematic under-provision in low-income and minority tracts. Maharjan et al. [6] confirmed this nationally, showing transit-access gaps disproportionately affect underserved populations across American metros. The federal Justice40 Initiative, directing 40% of infrastructure benefits to disadvantaged communities, demands exactly the granular, neighborhood-level measurement tools this project builds [7].

2.2 Urban Walkability and Equity

The EPA National Walkability Index—incorporating intersection density, land-use mix, and destination proximity—is the standard neighborhood walkability measure [8]. Research consistently shows walkability is unevenly distributed, with communities of color inhabiting systematically less walkable neighborhoods [2]. Toolkits such as the 15-minute-city score [10] operationalize walkable access but stop short of integrating other mobility modes.

2.3 Bicycle Infrastructure Equity

Bicycle infrastructure is a recognized component of equitable, low-carbon mobility, yet provision is uneven: dedicated facilities are concentrated in higher-income areas, and bikeability indices remain fragmented [4, 9]. OpenStreetMap, accessed through the OSMnx library [15], has become the standard open source for extracting and analyzing bicycle networks at scale.

2.4 Electric-Vehicle Charging Equity

As fleets electrify, equitable access to public charging—especially for households without home charging—is an emerging equity concern. Lou et al. [3] analyzed 121 million U.S. households and found disadvantaged communities have 64% fewer public chargers per capita, a gap rising to 73% among renters. Chen and Li [11] found that 60–80% of U.S. census tracts have no public charging access at all, with extreme geographic concentration.

2.5 Composite Indicators in Public Policy

Composite indicators compress multiple variables into a single, communicable number; the OECD/JRC *Handbook on Constructing Composite Indicators* [16] is the canonical methodological reference, structuring framework definition, normalization, weighting, and robustness/sensitivity analysis. It stresses that normalization and weighting are value-laden and must be paired with sensitivity analysis—a requirement this study meets through its weighting-scheme and threshold tests (§3.7–§3.8).

2.6 Spatial Analysis Methods

Because mobility infrastructure is spatially structured, the analysis uses established spatial-statistical tools. Global and local Moran's I [17] test whether MES values cluster in space and locate significant low-low cold spots. K-means clustering with silhouette-based model selection [18] derives an interpretable typology of neighborhood mobility profiles.

2.7 Gap in Existing Tools

The EPA Smart Location Database, DOE AFDC locator, the Urban Institute SITE tool [12], and the Oliver Wyman Forum Urban Mobility Readiness Index [13] each address one dimension or one geographic scale. None combines all four mobility dimensions into a tract-level equity score. Pandey et al. [14] found that rising urbanization is associated with increasing infrastructure inequality, reinforcing the urgency of targeted, evidence-based intervention tools.

3. METHODOLOGY

3.1 Study Area

The unit of analysis is the U.S. Census tract (average 1,500–4,000 residents). Three structurally distinct cities were selected for Phase 1 to stress-test the method across mobility regimes (Table 1); Phase 2 will extend to Phoenix AZ, Atlanta GA, and Detroit MI after pipeline validation. Selection criteria follow the proposal: population above 300,000, complete GTFS feeds, demographic diversity, and variation in transit-investment history.

Table 1: Phase-1 study cities and Census FIPS codes.

City	State (FIPS)	County (FIPS)	Tracts	Rationale
Chicago	Illinois (17)	Cook (031)	1,332	Mature high-frequency transit; diverse
Houston	Texas (48)	Harris (201)	1,115	Auto-oriented, sprawling; large low-income pop.
Seattle	Washington (53)	King (033)	495	Progressive transit and EV investment

3.2 Mobility Equity Score (MES)

The MES is a composite index computed for each census tract from four equally-weighted sub-indices (25% each), each normalized to a 0–100 scale by min–max scaling *within each city* before aggregation. Higher scores indicate greater mobility access relative to city peers. Table 2 defines each sub-index.

Table 2: The four MES sub-indices.

Sub-index	Weight	Key metric(s)	Data source
Transit access	25%	Stop density; AM-peak frequency (trips/hr); service span within 0.5 mi of tract centroid	GTFS (transit.land / agency)
Walkability	25%	EPA National Walkability Index (intersection density, land-use mix, destination proximity)	EPA Smart Location DB
Bicycle infrastructure	25%	Bike-lane km per km^2 of tract area	OpenStreetMap via OSMnx
EV charging access	25%	Public charger density per 1,000 residents; distance to nearest DC-fast station	DOE AFDC / OpenStreetMap

The composite is the weighted sum MES $= \sum_k w_k \cdot s_k^{\text{norm}}$ with $\sum_k w_k = 1$ and a baseline of equal weights. Tracts scoring below the 25th percentile of the composite MES are flagged as *mobility deserts*; tracts below the 25th percentile in two or more sub-indices simultaneously are *compounded mobility deserts*. The MES is cross-referenced with Census ACS variables: median household income, percent non-white, percent zero-vehicle households, and percent renter-occupied units.

3.3 Data Sources

All data are freely available with no licensing restrictions (Table 3).

Table 3: Data sources and access.

Source	Variables	Access
GTFS (transit.land / agency; Mobility Database)	Stop locations, schedules, headways	Free / token
EPA Smart Location Database v3	National Walkability Index (block group)	Free download
OpenStreetMap / OSMnx	Bike lanes, paths; EV charging stations	Free Python library
DOE AFDC	EV charger locations, level, network	Free REST API (key)
US Census ACS 5-Year (2019–2023)	Income, race, vehicle ownership, tenure	Free Census API (key)
Census TIGER/Line (2023, 2019)	Tract & block-group boundaries	Free download

3.4 Data Collection and Preprocessing

All layers are reprojected to EPSG:4326 for storage and mapping and to USA Contiguous Albers Equal Area (EPSG:5070) for any length/area/distance computation, so metrics are in true metres. GEOIDs are zero-padded strings (11 digits for tracts, 12 for block groups) to guarantee exact joins. A key preprocessing problem is *geographic-vintage mismatch*: the EPA Smart Location Database is published on 2019-vintage block groups, whereas the study uses 2023 tracts (to align with the 2019–2023 ACS). A naive GEOID-prefix join loses tracts wherever boundaries were redrawn—about 57% of Houston's. The pipeline therefore performs *areal interpolation*: the National Walkability Index is joined to 2019 block-group polygons (a 100% GEOID match) and area-weighted onto the 2023 tracts, each tract's value being the area-weighted mean of its overlapping block-group pieces. Missing values are preserved, not imputed, so data gaps remain visible.

3.5 Sub-Index Computation

Transit. From GTFS `stops` and `stop_times`, AM-peak (07:00–09:00) departures per stop are converted to trips/hour and the daily service span is computed. Stops are spatially joined to a 0.5-mile buffer of each tract centroid, yielding stop density, mean frequency, and span, which are normalized and averaged. **Walkability.** Area-weighted National Walkability Index per tract (§3.4), normalized to 0–100. **Bicycle.** OSM edges are retained as dedicated infrastructure when the `highway` tag is cycleway/path/track or a positive `cycleway` tag is present; clipped to tracts, bike-lane km per km^2 is normalized to 0–100. **EV charging.** Public stations are spatially joined to tracts; chargers per 1,000 residents and inverted distance to the nearest DC-fast station are normalized and averaged. The authoritative source is the DOE/NREL AFDC; where that API is unreachable, the pipeline falls back to OSM `amenity=charging_station` points, a key-free lower-bound proxy.

3.6 Normalization and MES Construction

Each raw sub-index is rescaled by min–max normalization, $x_{\text{norm}} = \dfrac{x - x_{\min}}{x_{\max} - x_{\min}} \times 100$, computed within each city, and combined as the equally-weighted sum in §3.2. A robust percentile-rank variant is retained for sensitivity checks.

3.7 Weighting Methodology and Sensitivity Analysis

Because equal weighting is itself a value choice, three alternatives are computed and compared: *entropy* weighting (objective weights from each sub-index's dispersion), *PCA* weighting (first-principal-component loadings), and *AHP* weighting (expert pairwise judgement). For each scheme the MES and the equity correlations are recomputed, and the stability of the conclusions is reported [16].

3.8 Mobility-Desert Classification

A mobility desert is a tract below the city's 25th MES percentile; a compounded desert is below the 25th percentile in ≥ 2 sub-indices. Because the cut is arbitrary, a threshold sensitivity analysis recomputes the desert set at the 20th, 25th, and 30th percentiles and reports the Jaccard overlap with the baseline and the demographic profile of flagged tracts at each threshold.

3.9 Equity Correlation Analysis

Pearson correlations relate the MES (and each sub-index) to the four demographic variables, with two-sided p-values and 95% confidence intervals from the Fisher z-transformation, $z = \operatorname{arctanh}(r)$, $\mathrm{SE} = 1/\sqrt{n-3}$, $\mathrm{CI} = \tanh(z \pm 1.96\,\mathrm{SE})$. A negative MES–income or MES–percent-non-white correlation would indicate that more-disadvantaged neighborhoods receive less mobility infrastructure.

3.10 Spatial Autocorrelation

Global Moran's I tests spatial clustering of MES using queen-contiguity, row-standardized weights with 999 permutations; Local Moran's I (LISA) classifies each tract as high-high, low-low, high-low, low-high, or non-significant, mapping low-low clusters as mobility cold spots [17].

3.11 Cluster Analysis

K-means clustering on the four standardized sub-indices derives a typology of neighborhood mobility profiles; k is chosen by maximizing the mean silhouette score over $k \in \{2, \ldots, 7\}$, with the inertia (elbow) curve reported for validation [18]. Each cluster is labelled from its centroid profile.

3.12 Dashboard Development

The dashboard (Plotly Dash with Dash Bootstrap Components) provides a tract-level choropleth of the MES or any sub-index with desert filters; a click-through side panel (score, city percentile, sub-index bars versus the city median, auto-generated interpretation, and demographics); a city equity-gap comparison (median MES of the poorest versus richest income quintile); a ranked table of the most disadvantaged tracts; and CSV/PNG export.

3.13 Reproducibility and Implementation Status

All processing is in Python 3.11 (pandas, geopandas, numpy, scipy, scikit-learn, PySAL). Every stage is an independently-runnable module, raw downloads are cached, and a `pytest` suite covers normalization, MES construction, and the correlation statistics. At submission the pipeline runs end-to-end for all three Phase-1 cities on measured open data across every dimension (transit via GTFS for all three cities, including METRO Houston resolved through the Mobility Database API). Global Moran's I for the MES is 0.82–0.92 ($p \approx 0.001$) across the three cities, confirming strong spatial clustering of mobility access; the substantive equity findings are deferred to Chapter 4.

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